Quantifying with sequencing

What is quantification?

1. Correlate a **biological process** using DNA fragments.
2. Connect the **abundance** of fragments to the **rate** of the biological process.

What are a functional assays?

A **biological process** represents a **biological function**.

Thus the **abundance of fragments** ought to correlate with the **activity level of the function**.

There are ongoing debates on the range of validity of each approach.

We "force" the **biological function** to produce **DNA fragments** that we can sequence.

What do we call quantification processes?

Typically named as: **SOMETHING-Seq**

Examples: **ChIP-Seq**, **RNA-Seq**, **RAD-Seq**

Usually work by making an known biological mechanism to produce DNA fragments of some known properties.

Misconceptions

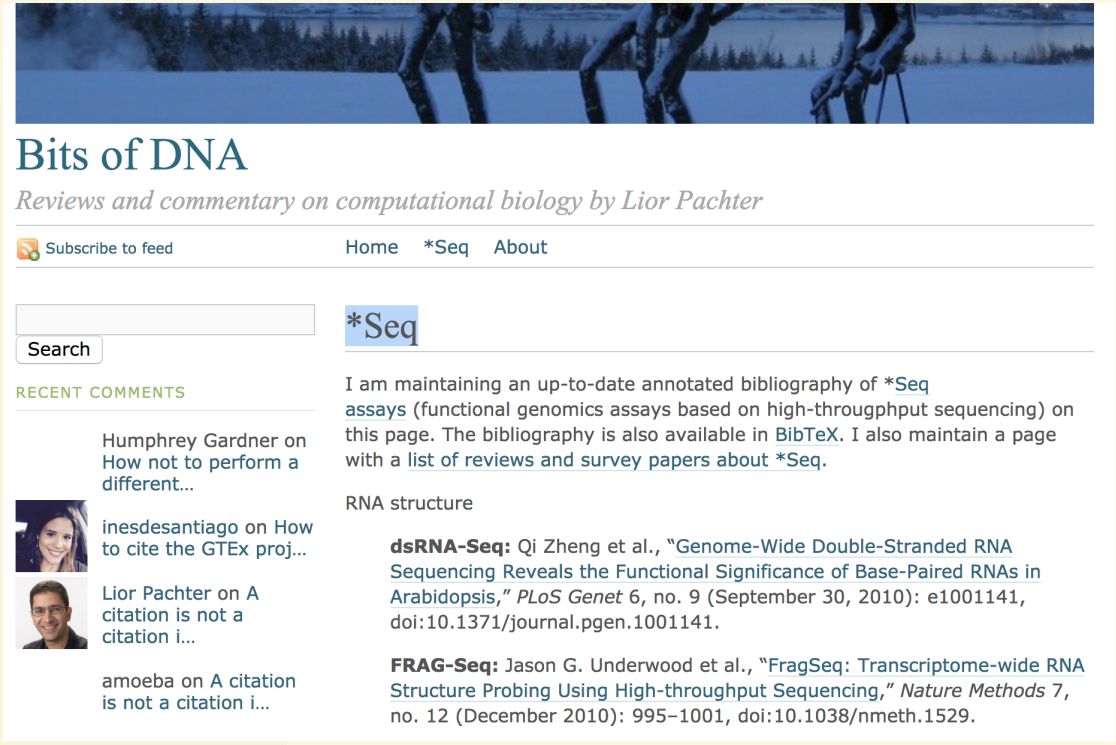
Common misinterpretions:

- *ChIP-Seq measures the rate of protein binding to DNA.*

Reality:

- *ChIP-Seq quantifies how frequent DNA fragments are. Bound DNA locations produce more fragments than unbound locations*

What types of Functional Assays exist?



Bits of DNA

Reviews and commentary on computational biology by Lior Pachter

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*Seq

I am maintaining an up-to-date annotated bibliography of *Seq assays (functional genomics assays based on high-througphput sequencing) on this page. The bibliography is also available in BibTeX. I also maintain a page with a list of reviews and survey papers about *Seq.

RNA structure

dsRNA-Seq: Qi Zheng et al., "Genome-Wide Double-Stranded RNA Sequencing Reveals the Functional Significance of Base-Paired RNAs in Arabidopsis," *PLoS Genet* 6, no. 9 (September 30, 2010): e1001141, doi:10.1371/journal.pgen.1001141.

FRAG-Seq: Jason G. Underwood et al., "FragSeq: Transcriptome-wide RNA Structure Probing Using High-throughput Sequencing," *Nature Methods* 7, no. 12 (December 2010): 995–1001, doi:10.1038/nmeth.1529.

Blog: **Bits of DNA**

Star-Seq --> *Seq maintained by Prof. Lior Pachter.

RNA structure

- dsRNA-Seq, FRAG-Seq, SHAPE-Seq, DMS-Seq

Chromatin structure

- ATAC-Seq, FAIRE-Seq, ChIA-PET-Seq, Nucleo-Seq

Transcription

- RNA-Seq, GRO-Seq, 3-Seq, TIF-Seq

Translation

- Ribo-Seq, Frac-Seq, GTI-Seq

Will the Functional Sequencing take over the world?

The way DNA "functions" is more important to understanding life then the way DNA "looks".

How do I analyze *Seq data?

1. What gets sequenced?

What kinds of fragments will be measured as reads.

2. What can quantified?

Which properties of the data correlate with the funcion.

Both can be suprisingly challenging get right.

For example for RNA-Seq

A cell has RNA of various types: `rRNA`, `mRNA`, `tRNA`, `snoRNA`, `miRNA`, etc.

- `mRNA` needs to be purified from total RNA.
- `RNA` needs to be reverse transcribed into DNA.
- Transcripts express at very different levels.
- Isoforms may be very similar
- Coverages may be very low
- Is the strand information preserved?

Each step introduces its own biases and challenges.

Quantification methods are similar

The analysis methods are typically very similar (perhaps with little twists).

1. Align
2. Quantify
3. Compare

Interpreting the results is different.

Understanding a `*Seq` method means understaning the origin of the DNA.

Base coverage vs Interval coverage

Base coverage

Base coverage is the numer of reads that "cover" a coordinate:

```
 ----   ------ -------
   -----   -----
              ----
01122112112222322111l0
```

Coverage: how many sequencing reads cross over a base.

Interval coverage



What is the coverage of transcript A? Is it

- `3`, `4` or `5`?

Depends on what "counting strategy" we use

`3` if all reads must be fully inside.
`4` if at least 50% of a read must be inside.
`5` if any overlap counts as a coverage.

*-Seq is all about coverages

1. How many reads cover something
2. Does that coverage change across conditions.

Compare across conditions

Same element different conditions:

1. Compute the coverage over **an interval** in condition **1** --> **100**
2. Compute the coverage over **the same interval** in condition **2** --> **200**

Does 100 -> 200 represent a statistically significant change?

Compare across intervals

Different elements under the same condition:

1. Compute the coverage over **an interval** in condition **1 --> 100**
2. Compute the coverage over **another interval** in the **same condition --> 200**

Does 100 -> 200 represent a statistically significant change?

RNA-Seq is a quantification

Most quantifications follow the workflow

1. Align
2. Quantify
3. Compare

There are many variations for each step.

Conceptually straightforward

Condition **1** --> **100** reads align to transcript **A**
Condition **2** --> **200** reads align to transcript **A**

Transcript A changes has a fold change of **2** --> **200/100**

Done.

Alas there are some complications

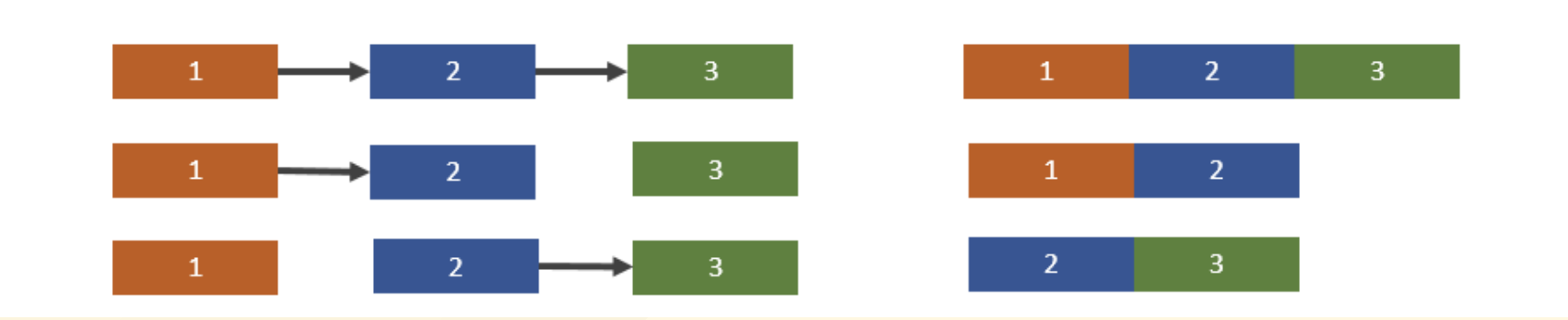
There always are

We need to deal with errors

RNA-Seq analysis is the process of accounting for all possible problems when making an otherwise trivial comparison

Isoforms

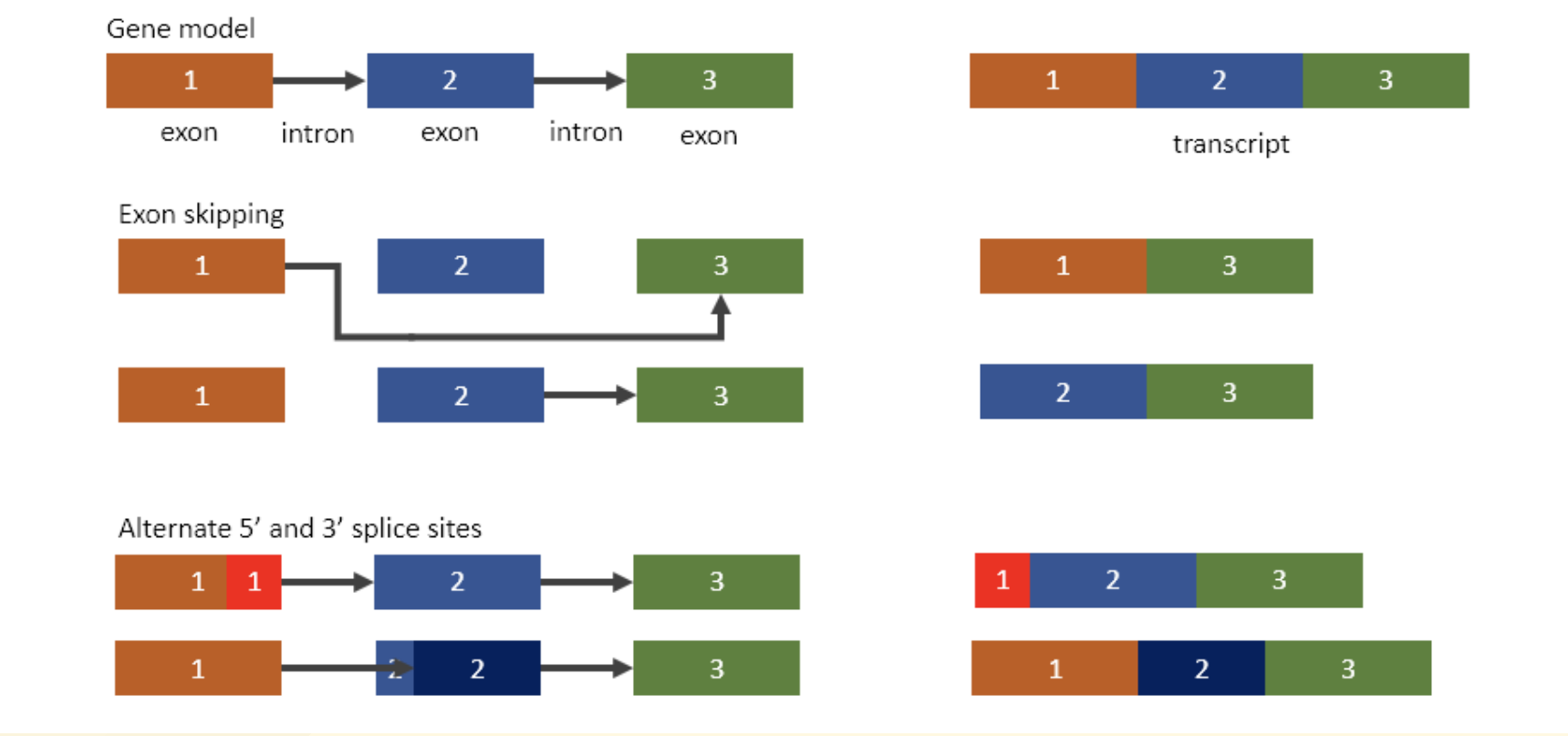
The same pieces of DNA may be assembled into different linear transcripts



You can detect transcripts by the reads that span over a change `1->2`, `2->3` etc.

Alternative splicing

The more common splicing mechanisms



RNA-Seq nomenclature

An RNA-Seq analysis has three separate yet equally important segments:

1. **Identify transcripts**
2. **Estimate abundances per transcript**
3. **Compare abundances --> differential expression**

There are workflows that mix and match from differnt methods.

1. Alignment Step

First choice in RNA-Seq analysis

Red pill or blue pill:

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1. Quantify against a genome

---

2. Quantify against a transcriptome

Think about advantages and disadvantages of each.

If you chose to use a genome as reference

1. It does not need an annotation.
2. It can discover novel transcripts.

---

1. Less accurate. It is more difficult to resolve ambigously alignments.
2. Non-expressed regions may affect the alignment.

If you chose a transcript as reference

1. Does not require a fully assembled accurate genome
2. Better quantification for similar transrcipt

---

1. Can't find novel transcripts
2. Requires good quality transcript information.

Should you quantify against transcripts or genome?

Do both!

What is gene level analysis?

Estimate abundance over the sum of all exons.

We pretend that there is just one long transcript, built from all the exons and call that the **gene**.

Sometimes it works - sometimes it does not. All depends on what is the origin of the phenomena under study.

Are gene level analyses valid?

In some cases yes.

When the phenotype is such that relative abundance of alternative trascripts don't matter.

When the phenotype is dominated by transcripts that belong to from different genes.

Data Replication

mRNA abundances is dynamic, it changes over time within a cell.

We need to ensure that any change we measure is due to the condition change and not the normal variability.

We need to make multiple measurements for the same condition.

To detect a change the variation across replicates has to be smaller than the variation between conditions.

Higher coverage or more replicates?

An evolving concept with tradeoffs.

The more data we collect the more accurate an individual estimate.

The more replicates we have the better we asses the natural variability.

Current recommendation:

More replicates are better: 4 or more.

2. Abundance estimation

How should we count an ambigous alignment?

This problem occurs only when aligning against a genome.

You may ahve reads that

1. Are not fully contained in a transcript
2. May align in multiple locations

Different counting (abundance estimation) strategies may produce different results.

It is worth testing other strategies

Counting is usually not that time consuming compared to other steps.

It is easier to re-run another strategy if the process is automated.

3. Abundance comparison

Comparing abundacnes

Normalization: the process of making data comparable.

The most common questions to be answered:

Is `Gene A` expressed in more copies than `Gene B` within `condition 1`?

Is `Gene A` expressed in more copies in `condition 1` vs `condition 2`?

A tortous history of abundance estimation

Suprisingly convoluted history (see Book for details):

1. Counts per million
2. RPKM: Reads per kilobase per million mapped reads
3. FPKM: like RPKM but for fragments
4. TPM: Transcripts per million
5. TMM: A statistical concept that estimates scale factors.

Each was designed to "protect" biologists from mathematics - only to end up being more difficult to

What should I use?

1. RPKM and TPM are ok to use to compare transcripts within a sample
2. Comparing across sample: TMM normalized counts

In class demo

We'll now follow the chapter

- **RNA-Seq: Griffith Test Data**

You will need R to be installed with proper packages.

See the chapter on **RNA-Seq Statistics.**